

University of South Australia

UniSA STEM

IT Projects

DECLARATION OF CONTRIBUTION

Project No: 3

The following is a declaration of your individual contributions towards this group assessment. If any contribution does not meet the assessment requirements, the course coordinator may adjust individual marks up or down, depending on the level of contribution made.

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I contributed 1310 words towards this assessment.

I worked on the following first 6 sections of the document.

Executive Summary, Project Purpose, Business Objectives, Project description, Project Objectives and success criteria, Requirements.

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I contributed 1205 words towards this assessment.

I worked on the following sections/questions (select whichever is appropriate)

1.3.3, 1.3.4, 1.3.5, 1.4, 1.5, 1.6

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1.7 Summary Budget, 1.8 Project Approval Requirements, 1.9.1 Project Manager Responsibilities, 1.9.2 Project Management Plan

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2.1 Use case analysis, 2.2 Functional requirements, 3 System Requirement Prioritisation.

Pepper the Robot Systems Requirements Document

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1. BUSINESS REQUIREMENTS

1.1 EXECUTIVE SUMMARY

The overarching goal of the project is to improve the functionalities of Pepper, an interactive humanoid robot, in order to create a more interactive, personalised and efficient client experience for clients within the clinic environment of Unisa Health Clinic. The key functionalities include: Enabling Pepper to dynamically greet clients, provide personalized conversations and relevant information to create a welcoming environment. Improve the feedback collect system and implement a user-friendly interface to manage and analyse the feedback. Enhance the Navigation and wayfinding functionality to enable Pepper to guide clients to the specific location, ensure the navigation is safe and efficient. Overall, the project will integrate advanced technologies to make Pepper more efficient and versatile in the clinic setting.

1.2 PROJECT PURPOSE/JUSTIFICATION

Unisa Health Clinic are keen to provide their great services to the communities and suburbs around the Unisa campus. Unisa Health Clinic would like to open new clinics in the North area to better serve the growing Northern communities. The project purpose is to explore and take advantage of the Pepper's full potential to provide a novel way to improve the client experience in the clinic setting.

1.2.1 Business Objectives

- Pepper can reduce the workload of the reception staff. Pepper can dynamically greet clients, provide personalised information and information about the clinic.
- Pepper can help Unisa Health Clinic to stand out against the other competitors.
- Pepper could be beneficial on feedback collection and provide actionable insight for clinic improvement.
- Pepper can securely and smoothly navigate clients to the allocated area to help clients get better experience.
- Pepper could be a better feedback collection channel, enabling clients to express their opinions and record it.
- Pepper can display the detailed information about the clinic to enhance the connection between clients and clinic.
- Pepper can help the reception desk staff to allocate clients to the waiting area, therapy room, check out area.
- Novel technologies integrated on the Pepper could attract more clients.

1.3 PROJECT DESCRIPTION

The project team has four team members all major in three year degree, Information Technology (Software Development). The time-bound for this project is 13 weeks, the team has 13 weeks to enhance the capabilities of Pepper the robot. To enable dynamic greeting, feedback collection, navigation, wayfinding and therapy sessions. The project team will ensure it is safe to deploy and run Pepper in the clinic environment. The project plans to start developing the greeting function first. At the same time, the project team will manage to find the algorithm for the navigation and wayfinding.

Once the greeting function is finished, the project team will start developing navigation and wayfinding functions.

1.3.1 Project Objectives and Success Criteria

Specific:

- Pepper the robot should be able to recognise patients coming into the clinic and greet the clients.
- Pepper the robot should be able to collect meaningful feedback from clients.
- Pepper the robot must be able to navigate the client to their allocated areas such as waiting area, therapy room, reception desk.
- When a task is finished, Pepper must return to the default spot and prepare for the next coming clients.
- During the therapy session, Pepper should be able to display the related video content on its screen.
- Pepper should always keep a safe distance with any people in the clinic environment to ensure everyone is safe.
- Pepper should stop when there is an obstacle in the way and only continue moving when the way is clear and safe to go.
- All the functionalities developed should be all encapsulated in a user-friendly interface in Pepper, so it can easily be accessed by anyone.

Measurable:

- The greeting function should be finished before week 6.
- The working demo of the navigation and wayfinding function should be delivered before week 9.
- The feedback system should be ready before week 11.
- Constantly test the development work and try to remove all the bugs to ensure a smooth running Pepper.

Attainable

- The project will utilise Choregraphe application to develop the function and behaviours of the Pepper and use Python language to develop any complicated algorithm needed for a Pepper robot behaviour.
- The project will utilise any useful existing libraries and function boxes in Choregraphe to accomplish the development goal.
- Demonstrate functional artefacts to project clients every fortnight.
- The project must get familiar with how to operate the Pepper such as moving the Pepper, charging, and installing applications on Pepper.

Realistic

- Pepper the robot will be deployed in the clinic environment to help Unisa Health Clinic provide better service to clients.
- Pepper the robot should reduce the workload of the staff in Unisa Health Clinic so that the staff can focus more on their medical procedures.
- Pepper the robot should enhance the connection between the Unisa Health Clinic and clients by offering an efficient feedback collection system.

Time-bound

- Total development time of this project is 13 weeks. The project team should deliver the refined work to the project before the project is closed.
- Background research and Requirements gathering should be conducted in the first 2 weeks.
- The SRS document should be completed in week 4.
- 5 sprint cycles should be conducted.
- Each developer of the project team should spend at least 10 hours on developing this project, cherishing the opportunity and time offered from the Project client. Ensure the final version delivered to the Project client meets all requirements.
- After each sprint cycle there should be a functional development work demonstrated to the Project client. Make sure the opinion and adjustment suggestion of Project client is recorded and fix them in the next sprint cycle.
- The Project consist of three stage:
 1. Research and SRS document, 3 weeks
 2. Project development, 9 weeks
 3. Project handover in last week, week 13.

1.3.2 Requirements

- Pepper the robot must dynamically greet clients, provide personalised conversations and relevant information to create a welcoming environment.
- The feedback collection system must be enabling clients to express their opinion about the service.
- Pepper the robot must have a user-friendly interface for staff to easily access the feedback data or any function the staff want to conduct.
- Pepper the robot must keep a safe distance to any people, existing clinic facilities in the clinic environment.
- Pepper must be able to stop when an obstacle blocks the way and only continue moving when the way is clear and safe to move.
- Pepper the robot must provide an efficient and safe navigation to the client when the client was allocated to a certain area.
- Pepper the robot must return to the default spot when a task is completed and prepare for the next coming clients.
- Pepper the robot should be able to distinguish the age, gender, expression of a client and determine what is the best way to interact with the client.
- Pepper the robot should be reliable and perform all the behaviours smoothly in the clinic environment.
- In the clinic environment, the body gesture and body movement of Pepper the robot must be appropriate to avoid causing any confusion in the clinic environment.
- Pepper the robot should facilitate the therapy process by providing video explanations related to the therapy.
- Pepper the robot should perform the feedback collection task after the clients finished the therapy.
- The user interface in the Pepper robot screen should clearly list what features it has and provide a brief description about the feature pre-selected.
- At the end of the project, a handbook should be delivered to the Project client. A training session for Unisa Health Clinic staff should be conducted if needed.

- The Development team should notify the Project clients when the navigation and wayfinding function is finished. Because the Project clients need time to arrange the safe transportation of Pepper the robot.
- Before testing Pepper the robot in the clinic environment, the project team should arrange a similar room to perform a mock testing session. To ensure all features are working as the project required, so that we can save the transportation cost and time.

1.3.3 Constraints

Throughout the project there will be multiple constraints that must be worked with, and around, to ensure our team is able to deliver the best product possible.

Time

There are many aspects of the project where time is a constraint. Due to Pepper being a task worked on over many semesters, by many teams, we must consider that our scope reflects that. By minimising the tasks outside of our navigation goal, we're able to manage our time a lot more efficiently than if we were juggling multiple tasks. This allows us to hand over a completed aspect of Pepper to the new team once our part is complete, opposed to parts of many incomplete projects, with the goal of the new team having the ability to pick straight off from where we've finished. To put this into practice we've created a Jira and will conduct weekly sprints with the idea of having something completed to present at the end of each.

Outside of the project timeline constraints, there are also time constraints that tie heavily into availability.

Availability

As all members of the team have alternate commitments, whether it be work, study, or family, availability arises to be a constraint. To work around this we've committed to scheduled meetups with all respective parties throughout the Pepper project. This meant creating spaces on relevant mediums that everyone felt comfortable with. We chose to create a Discord for the more casual discussion, and Microsoft Teams for the client side.

Prior Knowledge

As Pepper uses proprietary software that all members of our group are unfamiliar with, we've had to, and still continue to, learn how to interact within the space. The option to contact SoftBank directly has been explored, but so far deemed unnecessary.

Location

Pepper the robot requires various spaces to both program, and test, that would usually be out of our team's access permissions. We've been able to get permission to access the on-site location of Pepper, and will later move to the Health Clinic with Pepper to map out the premises.

1.3.4 Assumptions

While working on the project, the team has the following assumptions without being told:

- We all have the tools, or access to, to develop software.
- We all have the skills to develop software, or the ability to learn those throughout the task.
- We will be given generous On-Site Access to Pepper the Robot.
- We will be able to test Pepper in the health clinic within realistic limitations.
- We will maintain consistent communication reflective of what has been organised between the team, and clients.
- All parties will treat all members of the project with mutual respect.
- When help is needed all members will help where possible.
- Users of Pepper the Robot have the skills necessary to interact with technology similar to that of a tablet or phone.

1.3.5 Preliminary Scope Statement

The scope of our project will be to create a navigation system for Pepper the Robot. The majority of the project will be done within the Choregraphe software developed by SoftBank. A completed project will be a system that allows users to select from many destinations within the Health Clinic from the Pepper tablet, and Pepper will then guide them to the respective destination. Pepper will navigate obstacles throughout the clinic if present.

1.4 RISKS

Our team has identified many high-level risks that may influence the Pepper Robot project. The following is a RISK Matrix used to assess those risks.

Low (1)

Medium (2)

High (3)

Risk	Likelihood	Impact	Risk Rating (/9)	Response
Electrical hazards within the workspace	Medium	High	Six	We're able to mitigate, if not prevent this issue entirely, by inspecting the workspace thoroughly before developing. Once identified, electrical hazards will be dealt with by the appropriate parties.
Team member absent	High	Low	Three	By keeping each member up to date on what we've done, are doing, and intend to do,

				through verbal discussion and the Jira, we're able to each step up in the instance of an absent team member.
Data Breach of Pepper the Robot	Low	High	Three	We're able to eliminate the opportunities for a data breach of Pepper the Robot by developing in secure, known environments and taking care when in a new space
Pepper the Robot connecting to a malicious network	Low	High	Three	An external internet connection has been provided to the team to eliminate the risk of connecting to a malicious network.
Softbank unable to respond to an integral question	Medium	Low	Two	Softbank isn't the only medium to seek information on developing for the Pepper Robot. There are an array of online sources that can take the role in the instance Softbank doesn't reply.
The Pepper Robot may require a lengthy update halting development	Medium	Medium	Four	By both maintaining regular updates, and checking to see if there are any ahead of schedule, we can prevent the process delaying our development time. We're then able to schedule the updates for times where we're not working directly with the Pepper robot.
Our team fails to produce the code as intended	Low	High	Three	Continuous learning, and asking for help when needed are the only real ways to mitigate this risk.

The Pepper Robot is unable to navigate spontaneously placed objects	Medium	High	Six	The Pepper Robot code must be reflective of newly placed objects within a room opposed to relying strictly on a preset map.
Damage to the Pepper Robot	Low	High	Three	Our team has taken the time to learn how to safely handle Pepper. Alongside this, there will be no unauthorized users around Pepper that may not know how to handle the robot.
The Pepper Robot Hardware fails out of the team's control	Low	High	Three	In the instance that Pepper the robot malfunctions we must contact Softbank once it's deemed out of our fix.

1.5 PROJECT DELIVERABLES

On completion of the Pepper Project, our team will deliver what was identified to be the main aspects that the UNISA Health team requested.

1. A fully working navigation system for Pepper the robot. Pepper will have the ability to move from room to room around the health clinic.
2. A component for users to interact with Pepper the Robot, allowing them to select a room to move to.
3. A feedback component for Pepper the Robot, this will be used to collect useful data that will be passed onto the UNISA health clinic.
4. An overlay that allows users to exit the program and access relevant settings.
5. A brief user manual for the staff to train their use in Pepper the Robot.

1.6 SUMMARY MILESTONE SCHEDULE

Our team has put together an estimated schedule of all high-level project milestones. The estimated tasks may change as we further progress development.

Week	Estimated Tasks
1	Systems Requirement Document Set up workspaces
2	Systems Requirement Document Research Phase (This includes both how Pepper operates, our approach, and learning to code in Choregraph)

3	Develop Navigation Component
4	Develop Navigation Component
5	Develop Navigation Component
6	Develop Navigation Component
7	Finalise Navigation Component
8	Develop User Options Component
9	Finalise User Options Component Develop Overlay Component
10	Finalise Overlay Component Develop Feedback Component
11	Finalise Feedback Component Develop user manual
12	Testing of Pepper within the health clinic
13	Product Review Final Product Presentation

1.7 SUMMARY BUDGET

The reason why they stand prominently on Australia's list of highly sought-after roles is due to the substantial number of developers required to cater to the vast array of websites and digital services offered by contemporary businesses.

Australia has a robust tech sector, concentrated in key urban centers like Sydney, Melbourne, and Brisbane. The nation is experiencing a rising demand for skilled IT experts, particularly in the realm of software engineering. Both startups and established enterprises offer a range of employment opportunities.

The foreseeable expansion of software development careers in Australia remains promising. Given the continuous evolution of software development as a profession, there is a significant projected growth trajectory until 2026, solidifying it as a stable career choice.

This increasing demand for software developers is reflected in their substantial earnings in Australia, as evidenced by the latest job listings on SEEK, which offer salaries ranging from \$100,000 to \$130,000, inclusive of superannuation, contingent upon experience.

Conclusion, as per information from Google search, junior developers typically earn an average of \$82,750 – 84,458 per year or \$43.31 per hour. For this specific project, having four junior developers on board, each working for 10 hours per week over a span of 13 weeks, so it is $13 * 10 * 4 * 43.31$, would amount to a total cost of \$22,521.20. It's worth noting that since this project doesn't involve paid labor, this cost wouldn't be relevant. We don't care about it too much, but we can know and

need to know a rough summary budget. Some developers and senior developers are getting wages much higher than junior developers. Just take it as a reference. By doing this, we can know that our project team members can get wages a rough \$5630.30 per person replacing junior developers.

1.8 PROJECT APPROVAL REQUIREMENTS

- Recognizing the individuals and groups impacted by the project and actively involving them to comprehend their interests and requirements.
- After identifying these stakeholders, crafting a project plan becomes crucial (This plan should clearly detail the project's objectives, timelines, budget, and necessary resources).
- The prerequisites for obtaining approval for the project include:
 - Clearly defined objectives that align with the organization's strategic goals, particularly aiming to improve patient care and engagement.
 - A well-defined scope outlining the functionalities of the Pepper Robot, encompassing its ability to interact with patients, navigate autonomously, and handle patient data securely.
 - A comprehensive plan detailing crucial milestones, timelines, budget, and strategies for managing risks.
 - Adequate resources—financial, human, and technological—to support plan execution, including the necessary hardware and software for the Pepper Robot.
 - An evaluation process to gauge project success against objectives, involving consistent input from both patients and staff.
 - A governance framework establishing the roles and responsibilities of the project team, sponsor, and other stakeholders.
 - A capable team possessing the necessary skills and expertise to proficiently design, develop, and implement the Pepper Robot.

1.9 PROJECT MANAGEMENT

1.9.1 Project Manager Responsibilities

1. Ensuring the team remains on course and comprehends the project's prerequisites.
2. Offer any necessary resources vital for the project's triumph.
3. Lead the teammates and record what we work and what we have done and what time we have done.
4. Continuously plan and monitor the progress of all team members.
5. Prepare and maintain project, stage and exception plans as required.
6. Oversee project milestones such as arranging work segments, allocate tasks/user stories.
7. Sustain consistent communication between the client and the development team.

8. Supervise and strategize a project encompassing timelines, resource needs, and objectives. This entails harmonizing projects with the team, stakeholders, and clinic personnel for unified understanding.

9. Recognize potential risks and devise strategies to mitigate the consequences of unforeseen challenges on the project.

1.9.2 Project Management Plan

The project manager develops the project management plan based on inputs from the project team and key stakeholders. This plan is an official and approved document that outlines how the project will be executed, monitored, and controlled. It can range from a concise overview to a detailed account, encompassing baselines, subsidiary management plans, and other planning documents. This document serves to establish the methodology the project team adopts to accomplish the project management scope as intended.

Significant effort from project managers is dedicated to ensuring the accomplishment of baselines. This diligence guarantees that the project sponsor and the organization extract the complete advantages from their projects. Alongside effective initial planning, a project manager's competence lies in adeptly overseeing the project, ensuring timely delivery of project deliverables, and adhering to the project management plan.

As the project progresses, its performance is evaluated against the performance measurement baseline outlined in the project management plan. The performance measurement baseline consists of the scope, schedule, and cost baselines. If there are deviations from these baselines during the project's execution, the project manager addresses them by making necessary adjustments. In cases where these adjustments fail to rectify the deviations, formal change requests to the baselines are required.

Project managers invest a significant amount of time in ensuring that these baselines are met, which ensures that both the project sponsor and the organization reap the complete benefits of the project. Effective project management involves not only thorough planning but also proficient control of the project, ensuring that deliverables are timely and the project aligns with the project management plan.

The product manager will utilize the time management tool, Jira, to schedule sprints and establish deadlines for allocated tasks of team members. Regular visits to the kanban board will be made to identify any lagging tasks and to ensure the entire team is informed about such issues. During sprint reviews, the product manager will present the project's backlogs and roadmaps to the client, thereby maintaining them informed throughout the development phases.

Consistent communication with stakeholders will be upheld throughout the project's workflow. Weekly status meetings will be held with the project team, and clients will receive bi-weekly project updates. Any identified issues or risks will be openly discussed within the team and promptly communicated to the relevant stakeholders.

1.10 AUTHORISATION

Approved by the Project Sponsor: _____

Date: / /

Project Sponsor Name: _____

Project Sponsor Title: _____

2. SYSTEM REQUIREMENTS

2.1 USE CASE ANALYSIS

2.1.1 Use Case Scenario/Category 1 (Greetings)

1. As a client, I expect the Pepper robot to recognise someone is arriving and face them as soon as possible, so they know there is a Pepper robot coming to receive them.
2. As a client, I expect the Pepper robot to greet people kindly, with not only verbal greetings but also physical gestures to welcome them, so the patients will feel comfortable with the Pepper robot.
3. As a client, I expect the screen on the Pepper robot to have a basic button for users to press, which allows them to know some information they want to know, and the button function is straightforward and easy to use.
4. As a client, I expect the Pepper robot not only lets users press the buttons on the screen, but they also can talk to Pepper robot and get the same result as pressing the buttons, which may be easier for the elderly and children.
5. As a client, I expect the users can go back to the home screen or go back to the last step anytime they want, which in case there are misoperation or misunderstandings.
6. As a patient, I wish the Pepper robot to be able to communicate naturally with me and be able to make eye contact with me, which will make me feel comfortable and reliable.
7. As a patient, I wish the user interface of the Pepper robot is friendly and clear, which does not make me feel uncomfortable or even resistant.

2.1.2 Use Case Scenario/Category 2 (Navigation)

1. As a client, I expect the Pepper robot can ask the users where they want to go, which allows them to enter the destination not only by text but also by voice, that will help users save a lot of time to find where they want to go.
2. As a client, I expect the Pepper robot to know where the destination patients want to go and be able to take them there, so that it not only helps the patients find the way but also helps the clinic handle the patients more efficiently.
3. As a client, I expect the Pepper robot to know the way back after taking the patients to where they want to go.
4. As a client, I expect the Pepper robot to be able to recognise there are people coming when the Pepper robot is moving, then the robot will stop and wait for the other person to move out of its path, so it will make sure no one is at risk of collision.

5. As a client, I expect the Pepper robot to be able to kindly let patients know that the destination has arrived, by stopping in front of the destination and using gestures to guide the patients.
6. As a client, I expect users can stop the Pepper anytime they want in case they want to do something else.

2.1.3 Use Case Scenario/Category 3 (Feedback Collection)

1. As a client, I expect the Pepper robot to have a page which allows patients to leave some feedback, so that the clinic can improve itself based on that feedback.
2. As a client, I expect the feedback system to be more than simple good and bad choices, but to allow the patients to enter a short sentence or even a paragraph as feedback. So that the clinic can provide more specific information.
3. As a patient, I want the feedback system to be simple and clear so that it does not waste a lot of time.

2.1.4 Use Case Scenario/Category 4 (Therapy)

1. As a client, I would like the Pepper robot to have interactive games to help consult younger patients, so that will make the whole process easy and enjoyable.
2. As a client, I would like the Pepper robot to be able to translate for the foreign patients, which allows the foreign patients to get the same good treatment.

2.2 FUNCTIONAL REQUIREMENTS

2.2.1 Scenario/Category 1 (Greeting)

- R1: The Pepper robot must be able to recognise there are patients coming and turn its face to them.
- R2: The Pepper robot must be able to greet patients friendly through talking and body language.
- R3: The Pepper robot must have buttons to let users press on the home screen which have further features, and the buttons are not only by pressing by hand but also by voice.
- R4: The Pepper robot must be able to go back after any actions.
- R5: The Pepper robot must allow users to go back to the home screen anytime they want.
- R6: All the words must be clear, and all the buttons must be easy to press.

2.2.2 Scenario/Category 2 (Navigation)

- R1: The Pepper robot must be able to ask patients where they want to go.
- R2: The Pepper robot must be able to lead the patients to the place where they want to go.
- R3: The Pepper robot must be able to go back to the original place after leading the way.
- R4: The Pepper robot must be able to stop when someone is in its moving path.
- R5: The Pepper robot must be able to make a gesture to inform the patients that it has reached its destination.
- R6: The Pepper robot must be able to stop anytime the patient asks it to stop.

2.2.3 Scenario/Category 3 (Feedback Collection)

- R1: The Pepper robot must have 5 buttons to let users press, each one of the buttons means how the users feel about the experience in the clinic.
- R2: The Pepper robot must allow users to enter some words as feedback.

2.2.4 Scenario/Category 4 (Therapy)

- R1: The Pepper robot must have interactive games in it.
- R2: The Pepper robot must be able to translate.

3. System Requirement Prioritisation

The way our team prioritizes those system requirements is based on the Return-on-Investment (ROI) of each system requirement. (ROI = Business Value/Effort Value)

No.	User Story	Business Value	Effort Value	ROI
Category 1 (Greetings)				
7.	As a patient, I wish the user interface of the Pepper robot is friendly and clear, which does not make me feel uncomfortable or even resistant.	9	1	9
6.	As a patient, I wish the Pepper robot to be able to communicate naturally with me and be able to make eye contact with me, which will make me feel comfortable and reliable	8	1	8
2.	As a client, I expect the Pepper robot to greet people kindly, with not only verbal greeting but also physical gestures to welcome them, so the patients will feel comfortable with the Pepper robot.	8	2	4
1.	As a client, I expect the Pepper robot to recognize someone is arriving and face them as soon as possible, so they know there is a Pepper robot coming to receive them.	6	2	3

3.	As a client, I expect the screen on the Pepper robot to have a basic button for users to press, which allows them to know some information they want to know, and the button function is straightforward and easy to use.	8	3	2.67
5.	As a client, I expect the users can go back to the home screen or go back to the last step anytime they want, which in case there are misoperation or misunderstandings.	8	4	2
4.	As a client, I expect the Pepper robot not only lets users press the buttons on the screen, but they also can talk to Pepper robot and get the same result as pressing the buttons, which may be easier for the elderly and children.	6	5	1.2
Category 2 (Navigation)				
4.	1. As a client, I expect the Pepper robot to be able to recognise there are people coming when the Pepper robot is moving, then the robot will stop and wait for the other person to move out of its path, so it will make sure no one is at risk of collision.	7	4	1.75

3.	As a client, I expect the Pepper robot to know the way back after taking the patients to where they want to go.	8	5	1.6
1.	As a client, I expect the Pepper robot can ask the users where they want to go, which allows them to enter the destination not only by text but also by voice, that will help users save a lot of time to find where they want to go.	7	5	1.4
2.	As a client, I expect the Pepper robot to know where the destination patients want to go and be able to take them there, so that it not only helps the patients find the way but also helps the clinic handle the patients more efficiently.	7	7	1
5.	As a client, I expect the Pepper robot to be able to kindly let patients know that the destination has arrived, by stopping in front of the destination and using gestures to guide the patients.	5	6	0.83
6.	As a client, I expect user can stop the Pepper anytime they wanted in case they want to do something else	4	7	0.57
Category 3 (Feedback Collection)				

3.	As a patient, I want the feedback system to be simple and clear so that it does not waste a lot of time.	8	2	4
1.	As a client, I expect the Pepper robot to have a page which allows patients to leave some feedback, so that the clinic can improve itself based on that feedback.	8	3	2.67
2.	As a client, I expect the feedback system to be more than simple good and bad choices, but to allow the patients to enter a short sentence or even a paragraph as feedback. So that the clinic can provide more specific information.	8	4	2
Category 4 (Therapy)				
2.	As a client, I would like the Pepper robot to be able to translate for the foreign patients, which allows the foreign patients to get the same good treatment.	4	6	0.67
1.	As a client, I would like the Pepper robot to have interactive games to help consult younger patients, so that will make the whole process easy and enjoyable.	4	8	0.5